

COMPONENTS

Vehicle Ethernet

Smart technology is helping developers streamline infotainment system development. For this, the LAN9360, a single chip Ethernet controller with embedded protocols, facilitates audio endpoint solutions for audio video bridging (AVB) and inter-connection of a vehicle's infotainment devices. It supports audio transmission, secure boot, and remote updates over Ethernet, requiring no software integration, and easy and quick audio device configuration.



Microchip Technology
www.microchip.com

SoC for vehicle monitoring

The R-Car V3H system-on-chip (SoC) delivers improved deep learning performance for smart camera applications. It includes integrated IP for convolutional neural network (CNN) processing up to overall 7.2 TOPS while maintaining low-power consumption levels. The R-Car V3H supports perception stack, sensor fusion with radar and/or Lidar, and ISP with up to 8MP camera support, allowing quicker time-to-market with lower bill of material (BOM) cost.



Renesas Electronics Corporation
www.renesas.com

Hybrid IGBT family

With superior switching frequencies and reduced switching losses, the 650V CoolSiC Hybrid IGBT series is suitable for DC-DC power converters and power factor correction (PFC). The CoolSiC Hybrid IGBTs perform with significantly

reduced switching losses and increased switching frequency that allow a reduced passive component size and low bill-of-material cost. Hybrid IGBTs can improve electromagnetic compatibility and system reliability. Applications include battery chargers and inverters.

Infinite Technologies AG
www.infineon.com

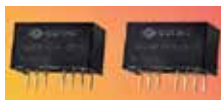
Battery monitoring IC

Designers of automotive battery management systems (BMS) can now deliver a high safety level for voltage, current, temperature, and communication while also saving space with the MAX17852 14-channel, high-voltage, automotive safety integrity level-D (ASIL-D) data-acquisition system. It is a small, cost-effective IC that offers high data accuracy with tight time-synchronisation, making it ideal for 48V and other automotive battery systems, which can see voltages up to 400V and beyond.

Maxim Integrated
www.maximintegrated.com

Isolated DC-DC converters

The new compact PQQ6W-S and PQQ10W-S regulator series offer cost-effective solutions for a wide range



of applications, including IoT, consumer, industrial, telecom and data communications. The PQQ6W-S provides up to 6W continuous power with up to 87 percent efficiency and 1,600V DC isolation, while the PQQ10W-S offers up to 10W isolated output with

up to 88 per cent efficiency and 1,500V DC isolation.

CUI Inc.
www.cui.com

5G modem

Supporting ultra-fast speeds on both non-standalone (NSA) and standalone (SA) architectures is the new M80



5G modem, which combines mmWave and sub-6GHz 5G technologies onto a single chip with a peak rate of 7.67Gbps in the downlink and 3.76Gbps in the uplink. The M80 supports dual 5G SIM, dual 5G NSA, and SA networks, and dual voice-over-new-radio (VoNR) for more reliable connectivity. Additionally, the M80 also offers support to ideal smartphones, PCs, Wi-Fi hotspots, broadband customer premise equipment (CPE), industrial IoT applications and more.

MediaTek
www.mediatek.com

Clock monitoring IC

New SmartClock features that facilitate monitoring of reference clocks have



been introduced in the Si5332-AM clock generators. They detect potential faults and provide built-in clock redundancy. On detection of a fault condition, SmartClock shares this information with an external microcontroller or system safety manager, which instructs the Si5332-AM to switch to a redundant backup source, ensuring continued and safe system operation.

Silicon Labs
www.silabs.com

Solid state relay

Offering circuit protection, power control, and sensing is the new CPC1561B